VICTORY 2027

SOUND

LECTURE-02 PHYSICS

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Topics To Be Covered

1. REVISION
2. REFLECTION OF SOUND WAVES
3. ECHO & REVERBERATION
4. QUESTIONS


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Revision

Sound → Mechanical Wave

Longitudinal Wave



$$v_s = \sqrt{\frac{E}{d}}$$

By Newton : $v_s = \sqrt{\frac{P}{d}}$

By Laplace : $\boxed{v_s = \sqrt{\frac{\gamma P}{d}}}$

$\gamma_{air} = 1.4$

$$\boxed{v_s \propto \frac{1}{\sqrt{d}}}$$

$$\boxed{v_s \propto \sqrt{T}}$$

$$\boxed{V_t = 332 + 0.61t}$$

$v_s \propto$ humidity

$\boxed{v_s \text{ independent of } P}$



REFLECTION OF SOUND WAVES

The return of a sound wave on striking a surface such as wall, metal sheet, plywood etc. back in the same medium is called the reflection of sound wave.

REFLECTION OF SOUND WAVES

LAWS OF REFLECTION

$\angle i = \angle r$

(i) The angle of reflection is equal to the angle of incidence,
(ii) The incident ray, reflected ray and normal at the point of incidence, all lie in one plane.

- Sound waves get reflected from any surface whether smooth or hard.
- The only requirement for the reflection of sound wave is that the **size of the reflecting surface must be bigger than the wavelength of the sound wave**.
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- The phenomenon of reflection of sound waves is utilized in making the megaphone (or speaking tube) and ear trumpet.







ECHO

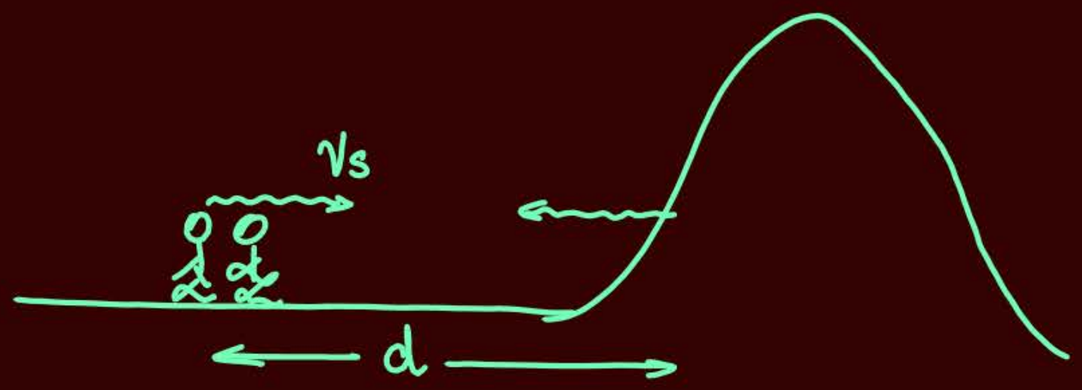
If a person stands at some distance from a wall (or a hillside) and produces a sharp sound, he hears two distinct sounds :

(i) The original (or direct) sound which is heard almost instantaneously and

(ii) The sound heard **after reflection from the wall** (or hillside) which is called an **echo**.

The sound heard after reflection from a distant obstacle (such as a cliff, a hillside, wall of a building, edge of a forest, etc.) after the original sound has ceased, is called an echo.

Note: The reflected sound if heard along with the original sound, is not the echo. Only the sound heard after the original sound has ceased (i.e., the sound distinctly separate from the original sound), is called the echo.



$$v_s = \frac{2d}{t}$$

$$340 = \frac{2d}{0.1}$$

$$d = \frac{34}{2} = 17\,m$$

At room temp (27°C)

$t < 0.1\,s$ (inaudible)

$t \geq 0.1\,s$ (audible)

$v_s = 340\ m/s$

$T \uparrow \quad v_s \uparrow \quad d \uparrow$

CONDITION FOR HEARING AN ECHO

An echo is heard only if the reflected sound reach the person at least 0.1 second after the original sound is heard. The reason is that the sensation of sound persists in our ears for about 0.1 second.

If d is the distance of the observer from the obstacle and V is the speed of sound in the medium. The time taken to hear the echo (or reflected sound) is

$$t = \frac{\text{total distance travelled}}{\text{speed of sound}} = \frac{2d}{\mathrm{V}}$$

$$d = \frac{\mathrm{V}t}{2}$$

By putting $t = 0.1s$ and $\mathrm{V} = 340\mathrm{ms}^{-1}$ in air at ordinary temperature,

$$\boldsymbol{d = \frac{340 \times 0.1}{2} = 17\mathrm{m}}$$

Thus, to hear an echo distinctly, the reflecting surface in air should be at a minimum distance of 17 m from the listener. **If the distance is less than 17 m , the reflected sound will reach the ears before the original sound dies out and so no echo will be heard.**

CONDITION FOR HEARING AN ECHO

Note:

(1) If the reflector is at a distance less than 17 m, the original sound just mixes up with the reflected sound.

(2) If there are repeated reflections at the reflecting surface, the sound gets prolonged. This effect is known as **reverberation** which can easily be experienced in tombs like Taj Mahal, Sikandra, etc.

(गूंज)

CONDITION FOR HEARING AN ECHO

Thus, to hear the echo distinctly, following three conditions must be satisfied:

(1) The minimum distance between the source of sound (or observer) and the reflector in air must be **17 m.**

- It is **different in different medium** depending upon the speed of sound in that medium.
- For example, inside sea water, $V = 1400 \text{ms}^{-1}$, so $d = \frac{1400 \times 0 \cdot 1}{2} = 70\text{m}$. i.e., to hear echo distinctly, the obstacle in sea water should be at a minimum distance of 70 m from the listener.

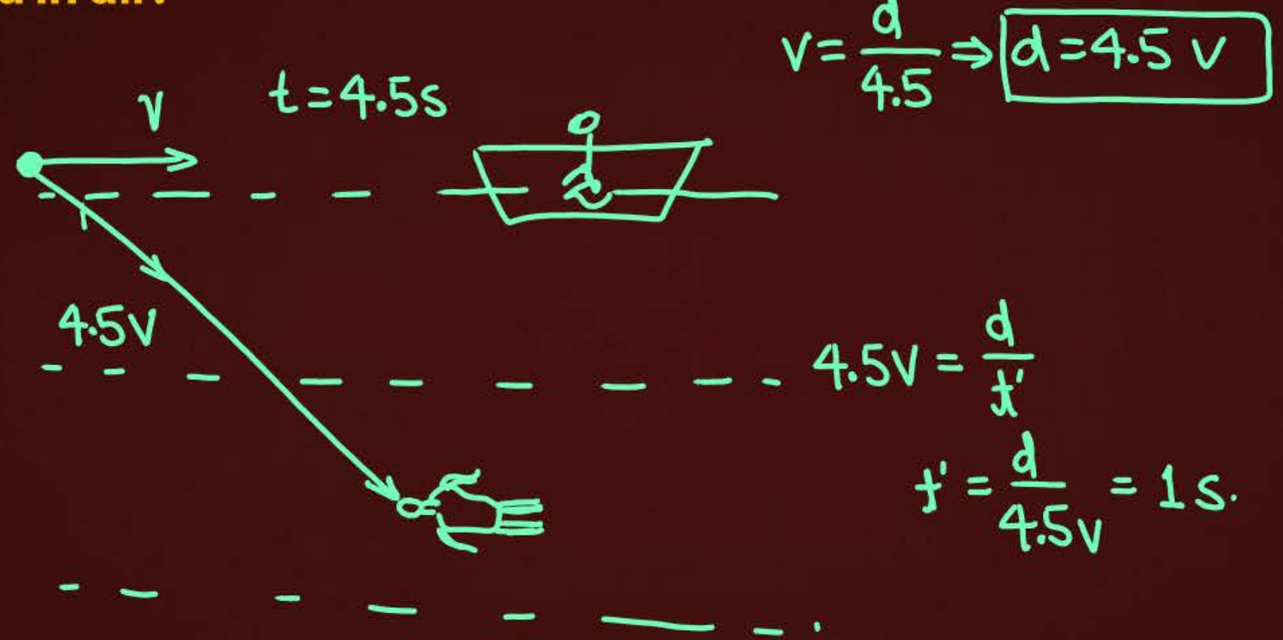
(2) The size of the reflector must be large enough as compared to the wavelength of the sound wave.

(3) The intensity of sound should be such that the reflected sound reaching the ear is sufficiently intense to be audible.

QUESTIONS

A sound produced on the surface of a lake takes 4.5 s to reach a boatman. How much time will it take to reach a diver inside water at the same distance if speed of sound in water is 4.5 times the speed of sound in air?

1. 1s
2. 2s
3. 3s
4. 4s



$v = \frac{d}{4.5} \Rightarrow \boxed{d = 4.5\,v}$

$4.5v = \frac{d}{t'}$

$t' = \frac{d}{4.5v} = 1\,s.$

QUESTIONS

A boy hears an echo of his own voice from a distant hill after one second. The speed of sound in air is 350 ms^{-1}. What is the distance of hill from the boy?

1. 125 m
2. 150 m
3. 175 m ✓
4. 200 m

$t = 1s$

$V_s = 350 m/s$

$V_s = \frac{2d}{t}$

$350 = \frac{2d}{1}$

$d = 175 m$



QUESTIONS

The minimum distance between the source and the reflector in air, to hear an echo, is approximately:

1. 10 m
2. 17 m
3. 34 m
4. 50 m



REVERBERATION

A sound created in a big hall will persist by repeated reflection from the walls until it is reduced to a value where it is no longer audible. The repeated reflection that results in this persistence of sound is called 'reverberation'.

Persistence of sound after the source has stopped as a result of repeated reflections from walls, ceilings, and other surfaces is called reverberation. Reverberation characteristics are important in the design of concert halls, theaters, etc.



QUESTIONS

A person clapped his hands near a cliff and heard the echo after 5 s. What is the distance of the cliff from the person if the speed of the sound, v is taken as 346 ms^{-1}?

1. 855 m
2. 865 m
3. 875 m
4. 885 m

$t = 5s$

$d = ?$

$v_s = 346\ m/s$

$$v_s = \frac{2d}{t}$$

$$346 = \frac{2d}{5}$$

$$d = 865\ m$$



USE OF ECHOES

SONAR

Max. Energy

fmax

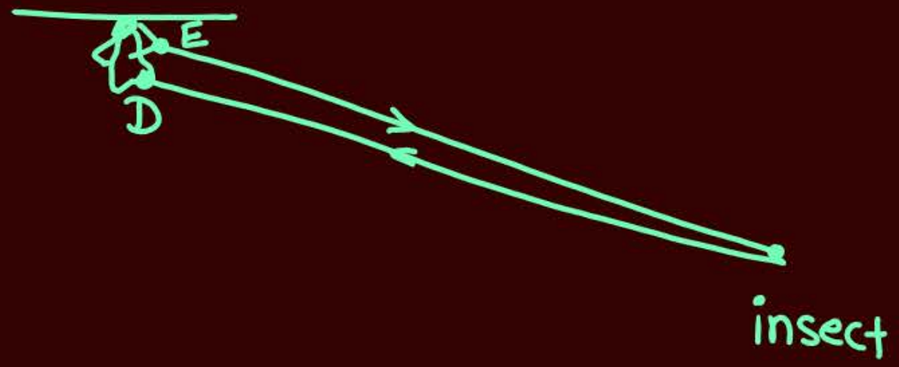
Echoes find their application in **sound ranging** and **echo depth sounding** by using the **ultrasonic waves**.

The ultrasonic waves (frequency above 20 kHz) are used because of the following three reasons :

(1) They can travel undeviated through a long distance,
(2) They can be confined to a narrow beam.
(3) They are not easily absorbed in a medium.

Note

Audible sound waves (frequency 20 Hz to 20 kHz) do not possess the above properties. However, the ultrasonic waves in a medium have the same speed as the speed of audible sound waves in that medium.


E
D
insect



USE OF ECHOES

1. USE OF ECHOES BY BATS, DOLPHINS AND FISHERMAN:

Bats can produce and detect the sound of very high frequency up to about 100 kHz. Bats fly with speed much lower than the speed of sound. The sounds produced by the flying bats get reflected back from an obstacle in front of it. By hearing the echo, bats come to know, even in the dark, the location of the obstacle, so they can fly safely without colliding with it. This process of detecting obstacle is called **sound ranging**.

Dolphins detect their enemy and obstacle by emitting the ultrasonic waves and hearing their echo. They use ultrasonic waves for hunting their prey.

A **trawlerman or fisherman** sends a pulse of ultrasonic waves from a source (a very high frequency vibrator) into the sea and receives the waves reflected from the shoal of fish in a detector.

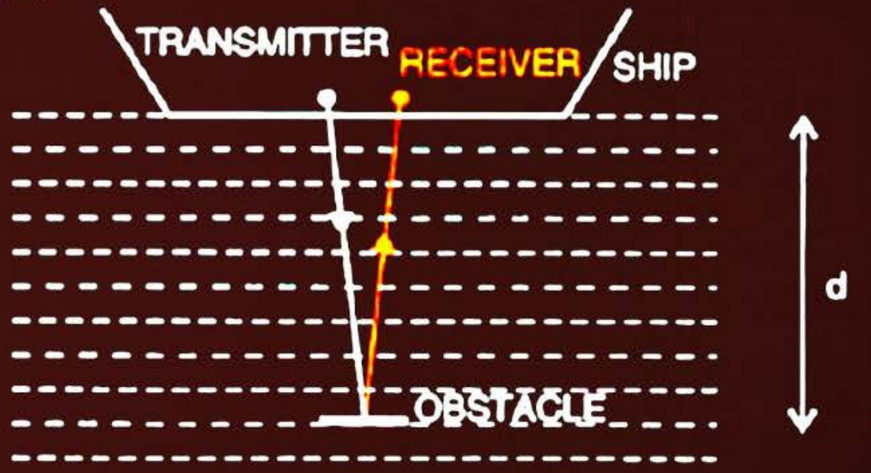
USE OF ECHOES

(2) USE OF ECHOES BY 'SONAR':

The word 'SONAR' stands for **sound navigation and ranging**. Fig. shows the principle of a sonar in which ultrasonic waves are sent in all directions from the ship. These waves are received after reflection from an obstacle such as the enemy submarine, iceberg, sunken ship, etc.

To find the distance of obstacle from the ship the time interval t between the instant when waves (pulse) are sent and the instant when waves are received, after reflection from the obstacle is measured.

The distance d of the obstacle from the source is then $d = \frac{Vt}{2}$ where V is the speed of ultrasonic waves in water. The depth of sea can also be found by this method. The process is then called **echo depth sounding.**



USE OF ECHOES

Note

(1) In **radar (radio detection and ranging)**, also the echo method is used to detect the presence of an obstacle and also to find its range. A signal of electromagnetic waves (such as radio waves or micro waves) is sent in space which after reflection from the object (such as enemy's aeroplane) in its path, returns to the radar itself.

(2) Both in 'SONAR' and 'RADAR', the transmitter and the receiver are placed close to each other.



USE OF ECHOES

(3) USE OF ECHOES IN MEDICAL FIELD:

- In medical field, echo method of ultrasonic waves is used for imaging the human organs (such as **liver, gall bladder, uterus, womb**, etc.) This is called **ultrasonography**.
- Similarly, **echo cardiography is used to obtain the image of human heart**.

QUESTIONS

If a sound wave is produced from a source and an echo is heard in 't' seconds afterwards, then if 'd' is the distance of the reflecting surface from the source and 'v' is the speed having wavelength λ and 'T' period of wave, then:

1. $d = \frac{2t}{\lambda}$
2. $d = \frac{\lambda t}{2T}$
3. $d = \frac{2T}{\lambda t}$
4. $d = \frac{\lambda T}{2t}$

λ, T

$v_s = \frac{2d}{t}$

$\frac{\lambda}{T} = \frac{2d}{t}$

$d = \frac{\lambda t}{2T}$

$V = f\lambda$

$V = \frac{\lambda}{T}$

QUESTIONS

A boy standing in front of a wall at a distance of 85 m produces 2 claps per second. He notices that the sound of his clapping coincides with the echo. The echo is heard only once when clapping is stopped. Calculate the speed of sound.

d

1. 320 m/s
2. 330 m/s
3. 340 m/s
4. 350 m/s

$$v_s = \frac{2d}{t}$$

$$= \frac{2 \times 85}{\left(\frac{1}{2}\right)}$$

$$= 340 \text{ m/s}$$

QUESTIONS

Imp

A boy stands 60 m (d) in front of a tall wall and claps. The boy continues to clap every time an echo is heard. Another boy finds that the time taken between the first and fifty-first clap is 18 s. Calculate the speed of sound.

Time of Echo = $\frac{18}{50}$ s.

2nd clap पर first echo सुनाई पड़ेगा।

51st clap पर 50th echo " "

$$v_s = \frac{2d}{t}$$

$$= \frac{2 \times 60}{\left(\frac{18}{50}\right)}$$

$$= 333.3 \text{ m/s}$$

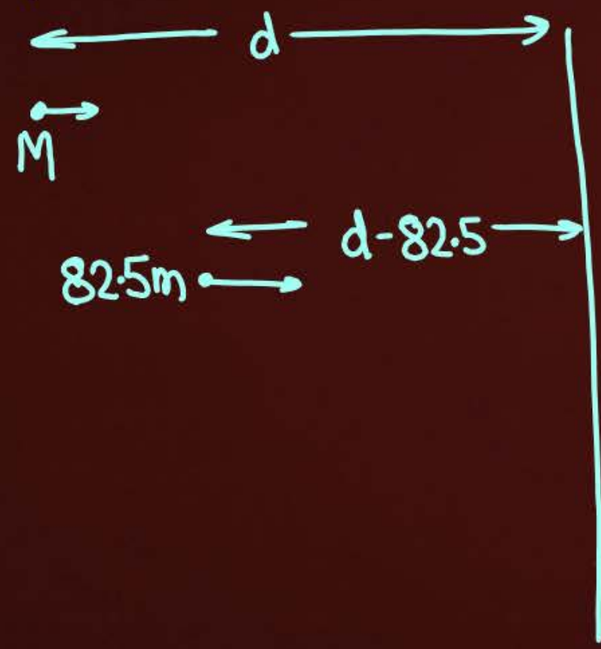
QUESTIONS

HW

A man standing in front of a vertical cliff fires a gun. He hears the echo after 3 s. On moving closer to the cliff by 82.5 m, he fires again and hears the echo after 2.5 s. Find:

(a) the distance of cliff from the initial position of man, and

(b) the speed of sound.



$$v_s = \frac{2d}{3} \quad \text{— (1)}$$

$$v_s = \frac{2(d-82.5)}{2.5} \quad \text{— (2)}$$

QUESTIONS

imp

A person standing between the two vertical cliffs and 640 m away from the nearest cliff, produces sound. He hears the first echo after 4 s and the second echo 3 s later. Calculate:

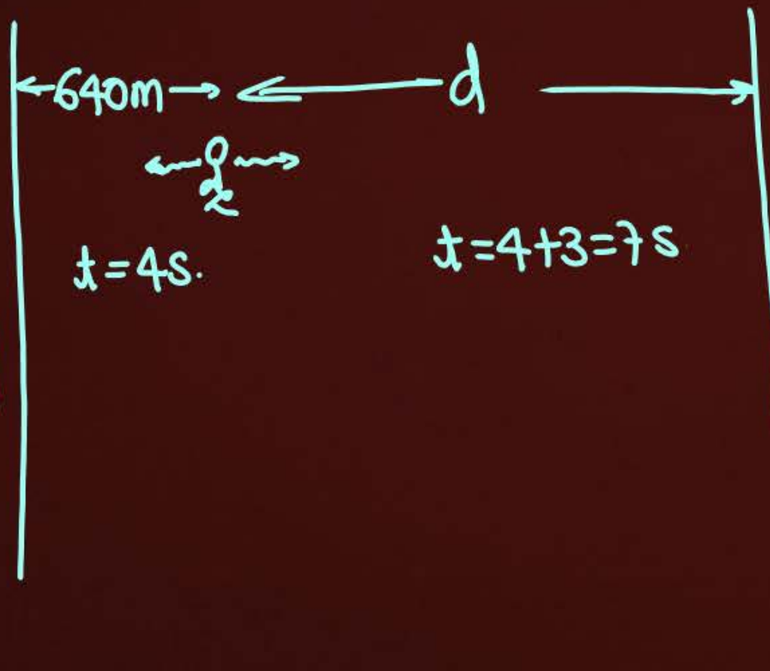
(a) the speed of sound in air, and

(b) the distance between the cliffs. 1760 m

$v_s = \frac{2d}{t}$

$= \frac{2 \times 640}{4}$

$v_s = 320\ m/s$



$v_s = \frac{2d}{t}$

$320 = \frac{2d}{7}$

$d = 1120\ m$



QUESTIONS

What does word SONAR stands for?

1. Sound navigation and ranging
2. Sound navigation and reaching
3. Sound network and ranging
4. Sound network and restore

HOMEWORK

- REVISE THIS LECTURE AGAIN.
- PRACTISE 20 QUESTIONS DAILY.
- COMMENT YOUR RESPONSE ABOUT THIS CLASS.



QUESTIONS

A pulse is sent from a ship to the floor of the ocean 420 m below the ship. 0.60 s later the reflected pulse is received at the ship. What is the speed of the sound in the water?

Solution:

Given, distance between source (ship) & reflector (ocean floor), $s = 420m$; time, $t = 0.6s$; speed of sound, $v = ?$

Now, speed,

$$v = \frac{2s}{t} = \frac{2 \times 420}{0.6} = \frac{840}{0.6} = 1400\text{m/s}$$

Thank You